MIL-B-20296
21 November 1951

MILITARY SPECIFICATION

BLANKS, ROTATING BAND; METAL POWDER COMPACTS

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-27-2
15 December 1943

This specification consists of this cover sheet and Specification 50-27-2, dated 15 December 1943, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer .

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BLANKS, ROTATING BAND; METAL POWDER COMPACTS

A. APPLICABLE SPECIFICATIONS.

A-1. Federal Specification QQ-M-151, General Specification for the Inspection of - of the issue in effect on date of invitation for bids, forms a part of this specification.

B. CLASS

B-1. This specification covers rotating band blanks made from powdered copper (see paragraph H-2).

C. MATERIAL AND WORKMANSHIP

C-1. Material. The blanks shall be made of copper powder of suitable quality and characteristics to meet all requirements specified hereinafter.

C-2. Workmanship.

C-2a. The blanks shall be in the form of a solid ring produced by compacting and sintering metal powder.

C-2b. All blanks shall be free from cracks, grooves, seams, excessive oxidation, flaws, deep scratches, and other defects which may render them unsuitable for the purpose intended. Minute voids normally will be present in the blanks, and the presence of such voids of normal size shall not constitute a basis for rejection provided that the blanks meet all other requirements of the specification.

C-2c. All blanks shall be free from oil, grease, and dirt.

D. GENERAL REQUIREMENTS

D-1. See detail requirements

E. DETAIL REQUIREMENTS

E-1. Chemical composition. The blanks shall contain not less than 99.00 percent copper, silver being counted as copper.

E-2. Density. The density of the blanks shall be not less than 8.0 grams per cubic centimeter.

E-3. Tensile properties. The tensile properties of test specimens made from the blanks shall be as follows:

Tensile strength, minimum, lb. per sq. in.	20,000
Elongation in 1 inch, minimum, percent	6.0

E-4. Bending. The blanks shall withstand the bending test described in paragraph F-7 without cracking on the outside of the bent portion.

E-5. Dimensions and permissible variations shall be as specified in the contract or purchase order, or shown on applicable drawings.

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F. METHODS OF SAMPLING, INSPECTION AND TESTS

F-1. Size of lots. A lot shall consist of not more than 20,000 blanks, of the same size, manufactured by the same processes, and from the same lot of metal powder.

F-2. Dimensions. At least fifty blanks from each lot shall be carefully checked for compliance with the dimensional requirements.

F-3. Samples for Government analysis. At least one sample for Government analysis shall be taken from each lot. Each sample shall be composed of drillings or millings taken from one blank selected by the inspector.

F-4. Samples for tests. At least 20 blanks shall be selected by the inspector from each lot. All sample blanks shall be tested for density. At least ten shall be subjected to the tension test and at least ten shall be subjected to the bending test.

F-5. Determination of density. Each sample blank shall be weighed on an analytical balance to the nearest 0.001 gram. Then the weight of the specimen suspended in water shall be determined to the nearest 0.001 gram. A few drops of a wetting agent such as "aerosol" shall be used to relieve the surface tension of the water. The blank shall be suspended by a fine silk thread. The water container shall be of glass and shall be supported in such a manner that the operation of the balance will not be affected. The density shall be calculated by the following formula:

$$\text{Density} = \frac{A}{A-B}$$

where:

A = weight of blank in air
B = weight of blank suspended in water.

F-6. Tension test.

F-6a. Specimen. The blank shall be cut open and flattened cold by pressure without hammering, and the tension test specimen shall be taken therefrom. The thickness and width of the specimen shall be the full thickness and width of the blank, provided that a small amount of metal shall be removed by machining if necessary to obtain a test specimen of uniform width and thickness in the test length.

F-6b. Tension tests shall be made in accordance with the applicable requirements of Specification QQ-M-151.

E-7. Bending test. The blank shall be flattened cold by pressure between parallel plates until the distance between the plates is four times the wall thickness of the blank. (i.e., until the distance between the inner walls of the blank is twice the wall thickness of the blank).

F-8. Retests. may be made in accordance with Specification QQ-M-151.

G. PACKING AND MARKING

G-1. Packing.

G-1a. Blanks shall be separated according to lot when packed for shipment.

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G-1b. Packing shall be such as to insure acceptance by common or other carriers for safe transportation, at the lowest rate, to the point of delivery.

G-2. Marking. Each shipment shall be legibly and indelibly marked to show the size, lot number, gross and net weight, quantity, name of the contractor, and the contract or purchase order number.

H. NOTES

H-1. Due to differences between characteristics of metal powder compacts and blanks made of wrought metal, the tests normally made on wrought bands after removal from the shell are not applicable to blanks covered by this specification, Where these blanks are applied to shell, appropriate tests should be incorporated in the shell specification or in the contract.

H-2. This specification is based on limited experience with 20 mm rotating bands and should be used only when specifically authorized.

H-3. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

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Note: Copies of this specification may be obtained from the Office, Chief of Ordnance, Army Service Forces, Washington, D. C.

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